Baseline Triage Model

Results, Metric Justification and Failure-Mode Reflection

Field	Detail
Project	AI-Assisted Triage - Mercer General Hospital
Phase	Phase 2 (baseline modelling), following the accepted Week 5 feasibility memo
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This report documents the first baseline model for the triage project, the initial classifier envisaged in the Week 4 pilot proposal, now that the Week 5 feasibility memo has been approved. Two interpretable baselines (**logistic regression** and a depth-bounded **decision tree**) predict ESI level from information available *at the point of triage* (vitals, age, arrival mode, chief-complaint flags). Categorical fields were one-hot encoded and numeric fields standardized inside a single ColumnTransformer / Pipeline, fitted on training data only. The data was split 80/20, stratified on Triage Level, with a fixed seed of 42. Post-triage and identity columns (esi, disposition, previousdispo, Unnamed: 0) were removed so the model only sees what a triage nurse would have at the moment of assessment. A stratified DummyClassifier provides a chance-level reference. As flagged since Week 5, the modelling data is a de-identified **US** academic-hospital dataset while the deployment target is a Caribbean ED so this is a proof-of-concept baseline, not a deployment-ready model.

1. Results

Overall performance (test set, n = 11,025)

Model	Accuracy	Macro F1	Weighted F1
Stratified Dummy (chance)	0.38	0.20	0.37
Logistic Regression	0.71	0.53	0.70
Decision Tree (max_depth=5)	0.56	0.29	0.51

Logistic regression reaches ~71% accuracy against a stratified-chance baseline of ~38% , the model comfortably beats a coin flip, satisfying Dr. De Fretias's bar for Phase 2. But accuracy hides the failure that matters, as the per-class figures show.

Per-class performance: Logistic Regression

ESI level	Precision	Recall	F1	Support
1 (critical)	0.40	0.25	0.31	16

ESI level	Precision	Recall	F1	Support
2	0.74	0.68	0.71	3,585
3	0.71	0.77	0.74	5,402
4	0.65	0.66	0.65	1,779
5	0.44	0.15	0.23	243

The model is strongest on the crowded middle band (ESI 2-3) and collapses on the rare extremes (ESI 1 and 5). For the most urgent class it correctly identifies only **4 of 16** critical patients. This is visible in the top row of the confusion matrix (w6_confusion_logreg.png): of 16 true Level 1 patients, 4 were flagged as Level 1, 11 were down-triaged to Level 2, and 1 to Level 3. For contrast, the stratified dummy catches **0 of 16** - so logistic regression is genuinely better than chance on the critical class, but a 25% catch rate is still far short of clinically acceptable. This directly confirms the class-imbalance concern surfaced in the Week 5 data profiling, where ESI 1 accounted for just 77 of 55,121 arrivals.

Macro vs weighted F1. Weighted F1 (~0.70) averages each class's score weighted by patient count, so the large, well-handled ESI 2-3 classes dominate and it looks healthy. Macro F1 (~0.53) averages all five classes equally, so the poor ESI 1 (0.31) and ESI 5 (0.23) scores pull it down. For a triage tool, macro F1 is the more honest headline - a rare class is not clinically unimportant just because it is small; ESI 1 is the *most* important, and weighted F1 lets the majority classes hide exactly the failure we care about.

2. Metric Justification

Primary metric: recall for ESI Level 1 - of the patients who were genuinely critical, the fraction the model successfully flags as critical.

On an imbalanced triage dataset, overall accuracy is dangerously misleading: because Level 1 patients are only ~0.14% of arrivals, a model that labels everyone "not critical" would still score ~99% accuracy while catching zero critical patients. The clinically meaningful question is not "how often is the model right?" but "of the patients who were truly in a life-threatening state, how many did the model catch?" We prioritise this over precision because the two failure modes are not symmetric: a false alarm (a stable patient flagged urgent) costs a few minutes of clinician attention, whereas a missed Level 1 patient can cost a life. This is not hypothetical, the Epic Sepsis Model analysed in our Week 4 proposal (Otles et al., 2021) failed in exactly this shape: strong aggregate performance masking poor detection of the cases that mattered, which then degraded clinical safety through alert fatigue. Choosing recall for ESI 1 over accuracy is the direct application of that lesson, so recall for ESI 1 not accuracy is the number this project should be judged on.

3. Failure-Mode Reflection

The failure I am most worried about is down-triage of ESI Level 1 patients. The model catches only 25% (4/16) of critical patients and routes the other twelve to lower-priority queues eleven to Level 2, one to Level 3. In a real ED this is the most dangerous error the system can make: a patient who needs immediate resuscitation is placed in a queue where they may wait, with no automated signal that they are deteriorating. The rarity of ESI 1 is both *why* the model struggles (few examples to learn from) and *why* the consequence is severe.

This is precisely why the Week 4 pilot design keeps the AI strictly non-autonomous. Under the **Friction Protocol**, the triage nurse commits an independent acuity score before any model output is shown, and the system can only ever advise never route. A 25% critical-catch rate confirms that the human-in-the-loop **Clinical Guardrail** is not optional polish but the core safety mechanism: the nurse's judgement, not the model, remains the safety net for exactly the Level 1 patients this baseline misses. Accordingly, this baseline is not safe to deploy for autonomous Level 1 detection; it makes the problem measurable, not solved.

4. Next Step

The immediate modelling priority is lifting ESI 1 recall via class weighting (class_weight='balanced') or resampling, and reporting ESI 1 recall as the headline KPI in every future model comparison so no later model can hide a poor critical-catch rate behind high overall accuracy. Beyond modelling, any move toward Grenada deployment must pass the Week 4 R2 governance gate quarterly re-validation against local Mercer patient cohorts with ≤2% performance variance because this baseline is trained on US data whose transferability to a Caribbean ED remains the project's central open caveat.

Reproducibility & Notes

Dataset yaleemmlc_admissionprediction_triage.csv; 80/20 split stratified on Triage_Level; random seed 42; Decision Tree max_depth=5; scikit-learn Pipeline (OneHotEncoder + StandardScaler → classifier). Figures generated from the corrected notebook (/notebooks); the fixed seed makes results reproducible. *An earlier pipeline accidentally left the target copy esi in the features, giving a misleading ~99.9% accuracy; removing it and the post-triage fields produced the honest results above.* Confusion matrix committed as docs/w6_confusion_logreg.png.